

New Research Study on Exploratory Testing based on the paper “Towards an Approach for Improving Exploratory Testing Tour Assignment based on Testers’ Profile”

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Exploratory Testing is a dynamic software testing approach that integrates test design, execution, and learning in real time. Unlike traditional scripted testing, which follows predefined steps, ET capitalizes on the tester’s creativity and real-time decision-making. ET not only contributes to thorough defect detection but also enhances software usability by simulating real user behaviors.

This paper proposes a research study aimed at optimizing ET outcomes in higher education by tailoring test tour assignments based on student profiles, building on insights from previous works, particularly the study "Towards an Approach for Improving Exploratory Testing Tour Assignment Based on Testers’ Profile."

The primary goal of this research is to investigate whether personalized test tour assignments, informed by students’ individual profile, can improve the effectiveness and efficiency of exploratory testing exercises in educational settings. Specifically, the study seeks to: assess how different student characteristics influence ET performance, evaluate the impact of tailored test tour assignments on defect detection rates and student engagement and compare the learning outcomes of personalized tour assignments.

Previous studies have demonstrated that ET, when incorporated into software testing education, leads to improved learning outcomes and testing proficiency. Evidence suggests that testers' backgrounds—including their personality traits, educational experience, and familiarity with software systems—may influence their effectiveness with certain types of test tours. Therefore, a more personalized approach may better support diverse student needs and lead to improved testing outcomes.

This study is inspired by prior research indicating that testers often gravitate toward tours perceived as easier or quicker, and that educational level correlates with tour preference. A structured investigation into these patterns within a university environment could inform the development of profile-based test tour recommendation systems.

The participants are undergraduate students enrolled in software testing or software engineering courses and the data collection uses the following:

- Pre-study survey: Assessing personality traits, learning styles, and software testing experience.
- Software testing task: Students will perform ET using a selected web-based application.
- ET session reports: Collected post-exercise for analysis of defect types, quantity, and severity.
- Post-study questionnaire: Gauging satisfaction, engagement, and perceived effectiveness.

The results will be analyzed by quantitative analysis of bug reports, correlation analysis between profile traits and performance metrics and qualitative content analysis of session reports and reflections.

This proposed study builds on existing research into exploratory testing and pedagogical methods in software testing education. Understanding the connection between student characteristics and ET performance can inform more effective teaching strategies, ultimately preparing students to meet the demands of software quality assurance.